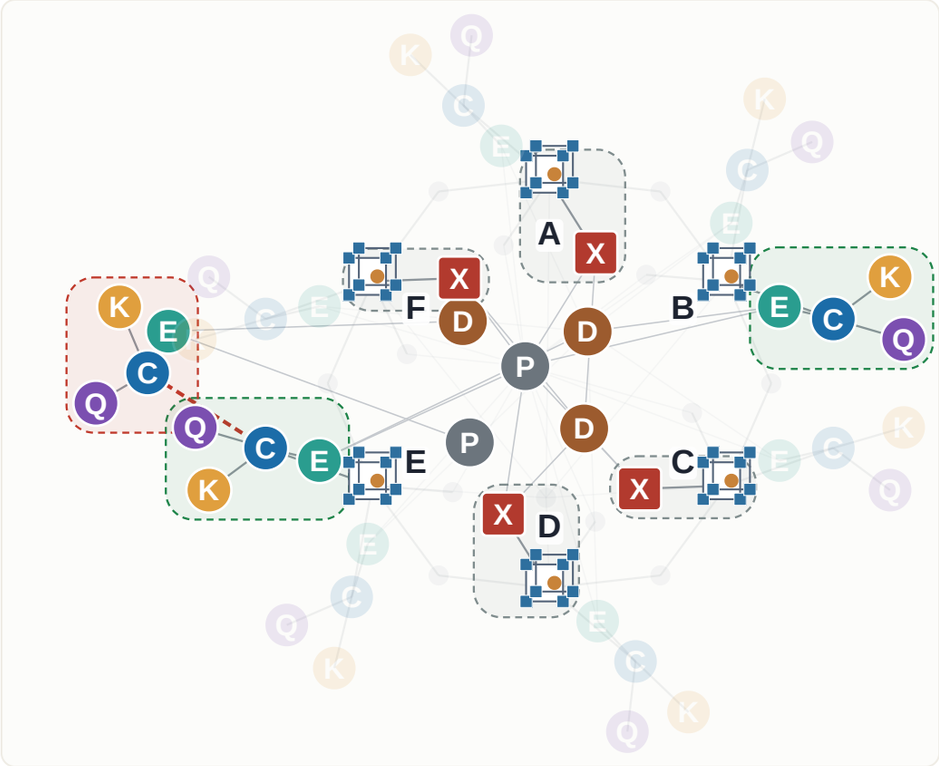
Question: “Which of MOFs A–F has the largest HSE06 band gap?” · **|N(q)|** = 64 nodes, **|J*|** = 27, **k** = 28

MOF	A	B	C	D	E	F
PBE gap (eV)	1.90	1.60	2.20	5.10	2.60	1.20
HSE06 gap in snapshot (eV)	none → X	2.85	none → X	none → X	4.70	none → X
ML prediction (HSE06)	—	—	—	—	3.12 (contradicts E)	—

a EPS (whole audit units)



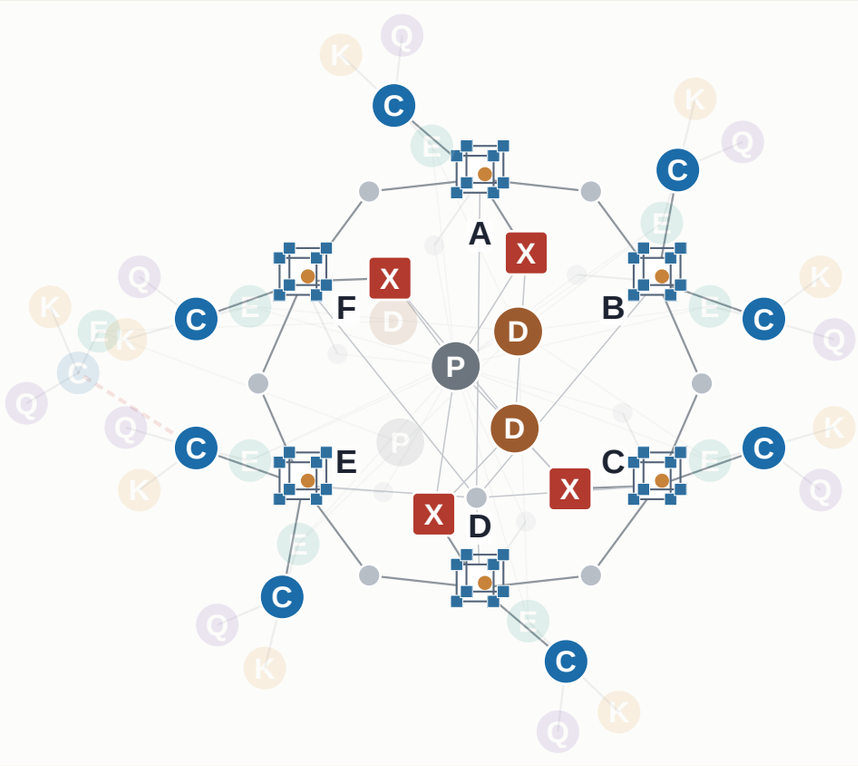
Conservative reader
MOF E: 4.70 eV (HSE06) ⚠ contested

Permissive reader
MOF E: 4.70 eV (HSE06) ⚠ contested

- ✓ Evidence & source (R1)
- ✓ DFT vs. ML (R3)
- ✓ Derivation / replay info (R5)
- ✓ Level of theory (R2)
- ✓ Contradictions (R4)
- ✓ Exclusions (R2x)

27 nodes kept · audit-sufficient: the answer can be replayed from the kept graph alone.

b Personalized PageRank (relevance), same budget



Conservative reader
Abstains (a kept claim has no condition or qualification node)

Permissive reader
MOF D: 5.10 eV (PBE)

- ✗ Evidence & source (R1)
- ✗ DFT vs. ML (R3)
- ✗ Derivation / replay info (R5)
- ✗ Level of theory (R2)
- ✗ Contradictions (R4)
- ✓ Exclusions (R2x)

28 nodes kept · neither correct nor auditable.

material (MOF) · claim · evidence · condition (level of theory) · qualification (DFT / ML) · provenance · derivation
exclusion record · context · contradiction · dashed boxes = audit units kept whole (support · contradiction · exclusion)